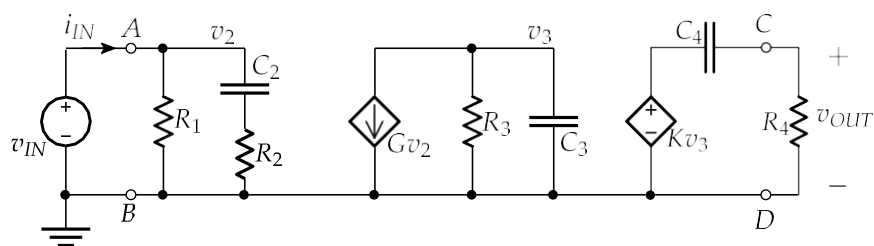


Homework 4

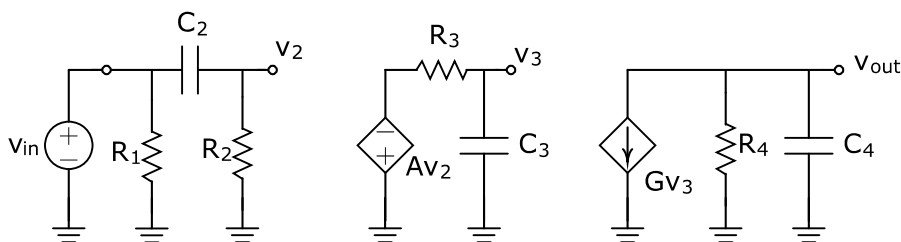
Topic: Frequency Response/Filters

1. For the following circuit,

- Find the analytical expression of the transfer function, v_{OUT}/v_{IN} , using only variables (do not plug in any numerical values)
- Sketch the Bode magnitude and phase plots using straight-line approximations and with $R_1 = 10 \text{ k}\Omega$, $R_2 = 37 \text{ k}\Omega$, $C_2 = 4.7 \text{ pF}$, $R_3 C_3 = 0.1 \text{ ms}$, $R_4 C_4 = 10 \text{ ms}$, $K = 10$ and $GR_3 = 10$.



2. For the following circuit



- What is the gain at low frequencies?
- What is the gain at high frequencies?
- Derive the expression for the transfer function $\frac{\bar{V}_{out}}{\bar{V}_{in}}$ and present it in the form that allows easy identification of poles and zeros.

3. For each of the straight-line approximation Bode plots below

- Draw a circuit that has a corresponding frequency response. The input resistance should not vary with frequency.
- Choose the values for the resistors and capacitors such that pole(s), zero(s), and transfer functions values match the frequency response shown.
- Write the expression for the transfer function for the filter circuit.
- What is the starting phase (ie., the phase at low frequencies)?
- What is the ending phase (ie., the phase at high frequencies)?

